ELECTRICAL MACHINES · MOCK

Machines mock paper

Practice questions · Day 23 · Sun 15 Nov · pack MX3

How to use this: this is a new paper, not a past paper. Sit it under exam conditions 2 days before the exam: 120 minutes, timer on, your printed notes and reference sheet only, calculator. Do not open the marking guide until time is up. Then mark yourself strictly with the guide and use the score table at the end to plan your last 24 hours.

Instructions

- Total duration: **120 minutes**.
- This examination is **open book**. Hardcopy printed and handwritten materials are allowed. No electronic devices apart from a calculator without remote communication.
- Attempt **ALL FOUR** questions. All questions carry equal marks (20 each, 80 in total). Part marks are given in brackets.
- Show your working. Method marks are given for correct equivalent circuits and correct substitutions, even when the final number is wrong.
- Phasors are rms. Unless stated otherwise, three-phase voltages are line-to-line and three-phase powers are totals.
- Answers should be expressed clearly and written legibly. These aspects of presentation will be taken into account in assessment.

Question	1a	1b	1c	2a	2b	2c	3a	3b	3c	4a	4b	4c	Total
Marks	5	6	9	7	5	8	6	5	9	5	7	8	**80**

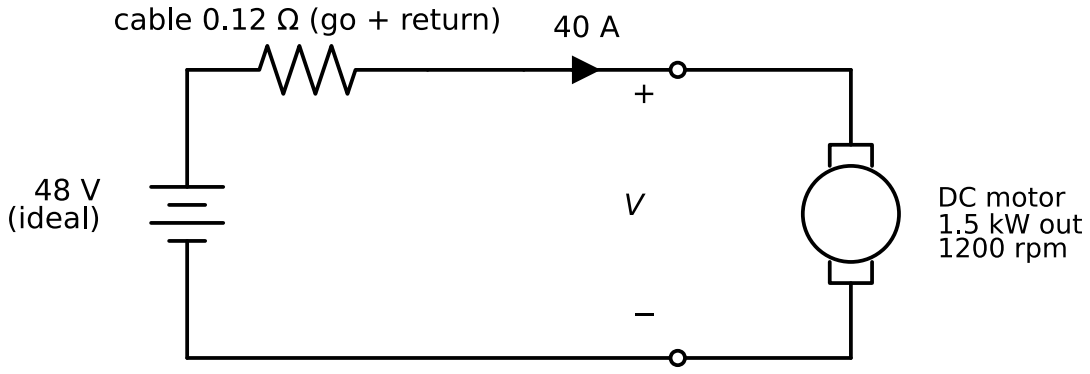
Time budget: 1.5 minutes per mark, so 30 minutes per question. Keep the last 5 minutes to check angles, signs and units.

Question 1 [20 marks]

AC and DC Power Circuit Analysis

1a) DC power and efficiency [5 marks]

A DC motor is supplied from an ideal 48 V DC source through a cable. The total resistance of the cable (go and return conductors together) is 0.12 Ω. The motor draws 40 A and delivers a mechanical output power of 1.5 kW at 1200 rpm.



Q1a: ideal 48 V source, 0.12 ohm total cable resistance, DC motor drawing 40 A

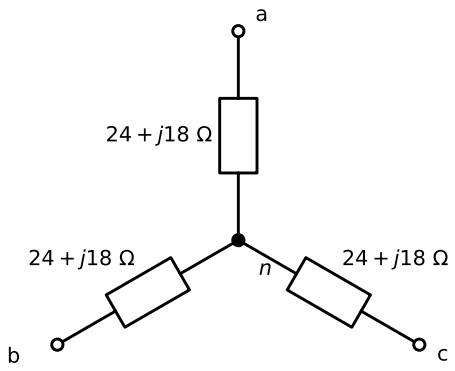
i. Find the voltage at the motor terminals and the power lost in the cable. [2 marks]

ii. Find the electrical input power to the motor, the motor efficiency and the motor output torque. [3 marks]

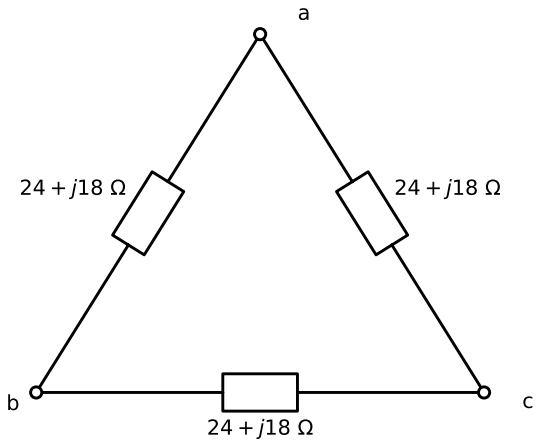
1b) Delta-connected load [6 marks]

A balanced three-phase load is connected in **delta** to a 415 V, 50 Hz, three-phase supply. Each phase of the load is a 24 Ω resistor in series with an inductive reactance of 18 Ω.

Star (part iii)



Delta (parts i, ii)



line voltage 415 V between any two of a, b, c

Q1b: three 24 + j18 ohm impedances in delta and then in star on a 415 V supply

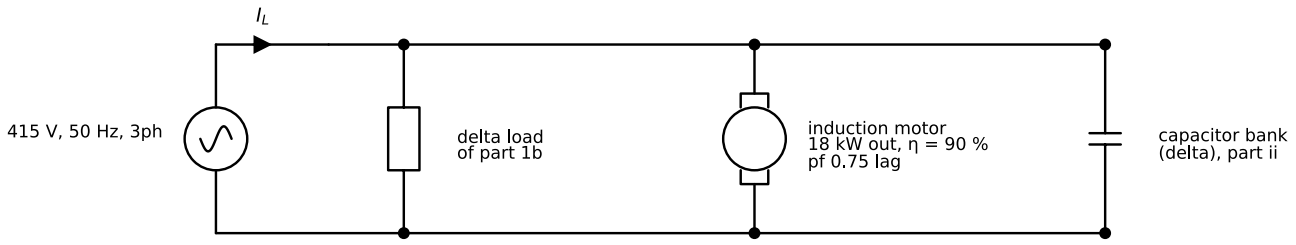
i. Find the phase current (magnitude and phase, with the phase voltage as reference), the line current magnitude and the power factor of the load. [3 marks]

ii. Find the total real power and the total reactive power absorbed by the load. [2 marks]

iii. The same three impedances are reconnected in **star** to the same supply. Find the new total real power. [1 mark]

1c) Power-factor correction [9 marks]

A small factory is supplied at 415 V, 50 Hz, three-phase. Its load consists of the delta-connected load from part 1b (connected in delta, as in parts (i) and (ii)) in parallel with a three-phase induction motor. The motor drives a mechanical load of 18 kW with an efficiency of 90 % and a power factor of 0.75 lagging.



Q1c: factory supply feeding the delta load, the induction motor and the delta capacitor bank in parallel

i. Find the total real power, the total reactive power, the power factor and the line current drawn from the supply. [3 marks]

ii. A bank of three capacitors is to be added in parallel with the factory to raise the supply power factor to 0.95 lagging. Find the total reactive power that the capacitor bank must absorb, and the new line current. [3 marks]

iii. The three capacitors are connected in **delta**. Find the required capacitance of each capacitor. [2 marks]

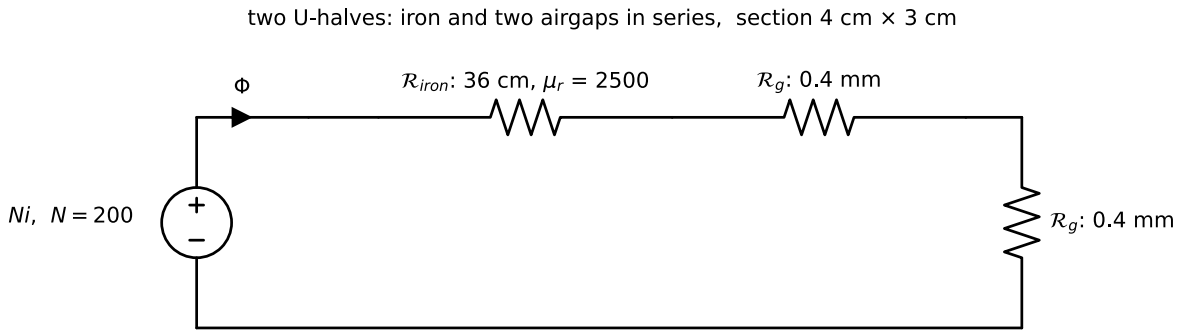
iv. How much reactive power would the capacitor bank need to absorb to give 0.95 **leading** instead? Give one reason why this is not done. [1 mark]

Question 2 [20 marks]

Magnetic Circuits and Transformers

2a) Electromagnetics [7 marks]

A magnetic core is made from two identical U-shaped halves placed with their ends facing each other, so that the flux crosses **two** airgaps in series, each of length 0.4 mm. The core has a rectangular cross-section of 4 cm × 3 cm everywhere, a mean magnetic path length in the iron of 36 cm (total for both halves), and a relative permeability of 2500. A coil of 200 turns is wound on one of the halves. Neglect fringing.



Q2a: magnetic circuit, MMF Ni with N = 200 driving flux through the iron and two 0.4 mm gaps in series

i. Calculate the reluctance of the iron, the reluctance of one airgap and the total reluctance, and hence the inductance of the coil. [3 marks]

ii. Find the DC coil current needed to produce a flux density of 1.2 T in the core, and the total energy stored in the two airgaps at this flux density. [2 marks]

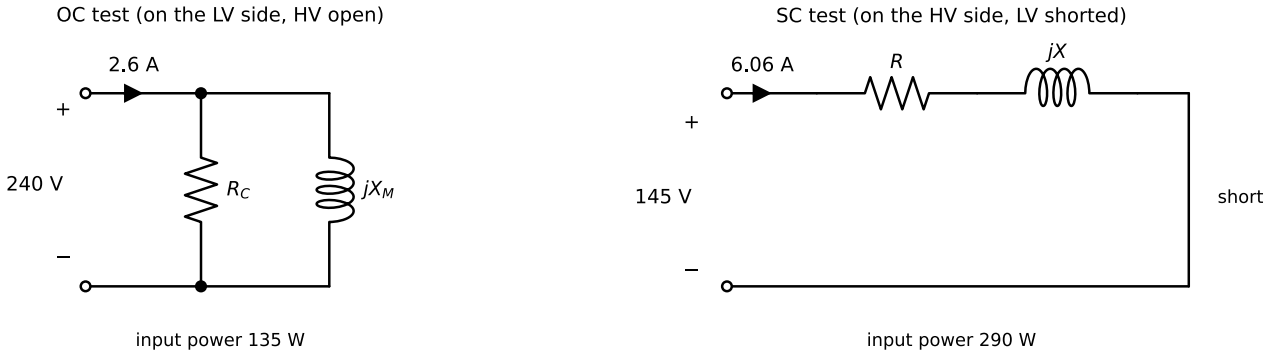
iii. At 1.2 T, find the total force pulling the two halves of the core together. [2 marks]

2b) Transformer parameters [5 marks]

The following tests are performed on a single-phase 3300 V : 240 V, 20 kVA, 50 Hz transformer.

- Open-circuit test, performed on the **low-voltage (240 V) side** with the HV side open: 240 V, 2.6 A, 135 W.
- Short-circuit test, performed on the **high-voltage (3300 V) side** with the LV side shorted: 145 V, 6.06 A, 290 W.

Find all the simplified equivalent circuit parameters (R, X, R_C and X_M), referred to the **high-voltage (primary)** side.



Q2b test circuits: OC on the LV side 240 V, 2.6 A, 135 W; SC on the HV side 145 V, 6.06 A, 290 W

2c) Transformer performance [8 marks]

The transformer of part 2b supplies its rated kVA at rated secondary voltage (240 V) to a load with a power factor of **0.8 leading**. Use the simplified equivalent circuit with the magnetising branch at the primary terminals, and take the referred secondary voltage as the phase reference.

Q2c: simplified circuit referred to the HV side with the 20 kVA, 0.8 leading load, V2′ as reference

i. Find the referred secondary current $\mathbf{I}_2'$ and the primary voltage $\mathbf{V}_1$ required, both in polar form. [3 marks]

ii. Find the voltage regulation in percent. Explain in one sentence why its sign is what it is. [2 marks]

iii. Find the efficiency at this load. [1 mark]

iv. Find the load (in kVA) at which this transformer has maximum efficiency, and the value of that maximum efficiency at 0.8 power factor. Assume rated voltage. [2 marks]

Question 3 [20 marks]

AC and Induction Machines

3a) AC machines [6 marks]

A three-phase AC machine has a nameplate showing 60 Hz and a rated speed of 1164 rpm.

i. How many poles does it have? Is it a synchronous machine or an induction machine? Briefly explain why. [2 marks]

ii. Find the rated slip and the frequency of the rotor currents at rated speed. [1 mark]

iii. At rated speed, what is the ideal (that is, maximum possible) efficiency of the machine? [1 mark]

iv. The machine is operating at one-half of rated torque. Assuming a linear torque versus speed curve, find the operating speed in rpm and the output power as a percentage of rated output power. [2 marks]

3b) Induction machine parameters [5 marks]

The following tests are performed on a 400 V, 50 Hz, three-phase induction motor whose stator is **delta-connected**. All voltages and currents below are **line** values as measured at the motor terminals, and all powers are three-phase totals.

- Stator resistance: 2.8 Ω per phase winding.
- No-load test: input voltage 400 V, input current 4.2 A, input power 320 W.
- Blocked-rotor test: input voltage 92 V, input current 10 A, input power 620 W.

Find the simplified equivalent circuit parameters per phase of the delta: R_2', X (the total leakage reactance $X_1 + X_2'$), R_C and X_M.

Q3b test circuits per phase of the delta: no-load 400 V, 4.2 A, 320 W; blocked rotor 92 V, 10 A, 620 W; R1 = 2.8 ohm

3c) Induction machine calculation [9 marks]

A 415 V, 50 Hz, six-pole, three-phase, **star-connected** squirrel-cage induction motor has the following equivalent circuit parameters per phase (referred to the stator where appropriate):

- stator resistance R_1 = 0.6 Ω, rotor resistance R_2' = 0.45 Ω
- stator leakage reactance X_1 = 1.2 Ω, rotor leakage reactance X_2' = 1.2 Ω
- magnetising reactance X_M = 40 Ω, core-loss resistance R_C = 300 Ω

Use the simplified equivalent circuit (magnetising branch across the input terminals). Friction and windage can be neglected. The motor runs at rated voltage at 970 rpm.

Q3c: 300 ohm parallel j40 ohm at the terminals, then 0.6 ohm, j1.2 ohm, j1.2 ohm and R2′/s

i. Find the synchronous speed, the slip and the effective rotor resistance R_2'/s. [2 marks]

ii. Find the referred rotor current $\mathbf{I}_2'$ and the input phase current $\mathbf{I}_1$ (magnitude and phase, with the phase voltage as reference), and the input power factor. [3 marks]

iii. Find the input power, the airgap power, the output torque and the efficiency. [3 marks]

iv. Find the slip and the speed in rpm at which maximum torque occurs. [1 mark]

Question 4 [20 marks]

Per-Unit and Synchronous Machines

4a) Synchronous machine tests [5 marks]

A 60 MVA, 13.8 kV, 50 Hz, three-phase, star-connected synchronous generator gives the following test results at a field current of 180 A:

- open-circuit line voltage: 9.2 kV
- short-circuit current: 1.25 kA

The stator resistance is 0.05 Ω per phase. Use rated values as the per-unit base.

Q4a test circuits: OC 9.2 kV line and SC 1.25 kA, both at a field current of 180 A, R = 0.05 ohm

OC and SC characteristics with the Q4a test point at 180 A

i. Find the base impedance and the base current. [1 mark]

ii. Find the synchronous reactance in ohms and in per-unit. [3 marks]

iii. What field current gives rated voltage on open circuit? Assume the open-circuit characteristic is linear. [1 mark]

4b) Generator at leading power factor [7 marks]

The generator of part 4a is connected to a large grid at rated terminal voltage. It delivers 45 MW at a power factor of **0.85 leading**. Neglect the stator resistance and saturation.

Q4b: generator E0 behind jXs at rated terminal voltage, delivering 45 MW at 0.85 pf leading

i. Find the terminal voltage and the stator current phasors in per-unit, with the terminal voltage as reference. Find also the stator current in kA. [2 marks]

ii. Find the excitation voltage E_0 in per-unit, the rotor angle, and the field current. [3 marks]

iii. With the field current held at this value, the operator tries to raise the real power output to 60 MW. Find the theoretical maximum power, in MW, that the generator can deliver without losing synchronism, and say whether 60 MW is possible. If it is not, find the minimum field current that would make 60 MW theoretically possible. [2 marks]

4c) Short-answer questions [8 marks]

Answer the following questions using words only (that is, without any sketches, drawings or equations).

i. A synchronous generator is connected to a large power grid. Explain what is meant by over-excited and under-excited operation, and explain how the field current and the mechanical input power each affect the real and reactive power delivered. [3 marks]

ii. The speed of an induction motor can be reduced either by adding resistance to the rotor circuit of a wound-rotor machine, or by using an inverter to reduce both the supply voltage and the frequency. Explain why the first method is inefficient and why the second is not. [3 marks]

iii. Explain why the per-unit system is useful when analysing a power system that contains transformers. [2 marks]

END OF QUESTIONS